

FORD GoBIKE

Exploratory Data Analysis Report

Executive Summary

This report summarises an exploratory data analysis (EDA) of Ford GoBike (now Bay Wheels) trip-level data for the San Francisco Bay Area bike-share system. The analysis covers 94,802 individual trips and examines when, where, and by whom bikes are used, in order to generate data-backed recommendations for fleet operations, station planning, and rider-segment marketing.

94,802 Total Trips Analysed	87% Trips by Subscribers	9.3 min Median Trip Duration	8-9 AM & 5-6 PM Peak Commute Hours
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The data shows a clear, commute-driven usage pattern. Subscribers generate the large majority of trips, ride for short durations concentrated around weekday rush hours, and are concentrated in the 25-40 age band. Customers ride less often but for longer durations, more evenly through the day, and are proportionally more active on weekends. A small set of stations near transit hubs and business districts account for a disproportionate share of both pick-ups and drop-offs.

Problem Statement & Business Objective

Problem Statement

Bike-sharing operators like Ford GoBike need to understand when, where, and by whom their bikes are being used in order to run an efficient and profitable service. Without this understanding, bikes end up unevenly distributed across stations, causing shortages at high-demand locations and surpluses at low-demand ones, while marketing and pricing decisions are made without evidence of actual rider behaviour.

This project analyses historical Ford GoBike trip data to answer questions such as: How does ridership vary by month, day of week, and hour of day? Which stations are busiest? How does trip duration differ

between Subscribers and Customers, and between genders? What does the rider age distribution look like?

Business Objective

- Improve bike and dock availability at the right stations at the right times.
- Identify the most valuable rider segments so marketing and subscription pricing can be tailored to them.
- Design strategies to convert casual Customers into recurring Subscribers, increasing both operational efficiency and recurring revenue.

Dataset Overview & Data Quality

Each row in the dataset represents a single trip and includes trip duration, start/end time, start/end station id, name and coordinates, a bike id, user type (Subscriber or Customer), member birth year, member gender, and (in later files) bike-share-for-all status. Missing values are concentrated mainly in member birth year and member gender, since Customers (casual, non-subscribed riders) are not required to provide this information. A small number of exact duplicate rows were found and removed. The data spans multiple months, enabling month-over-month and day-of-week seasonality analysis.

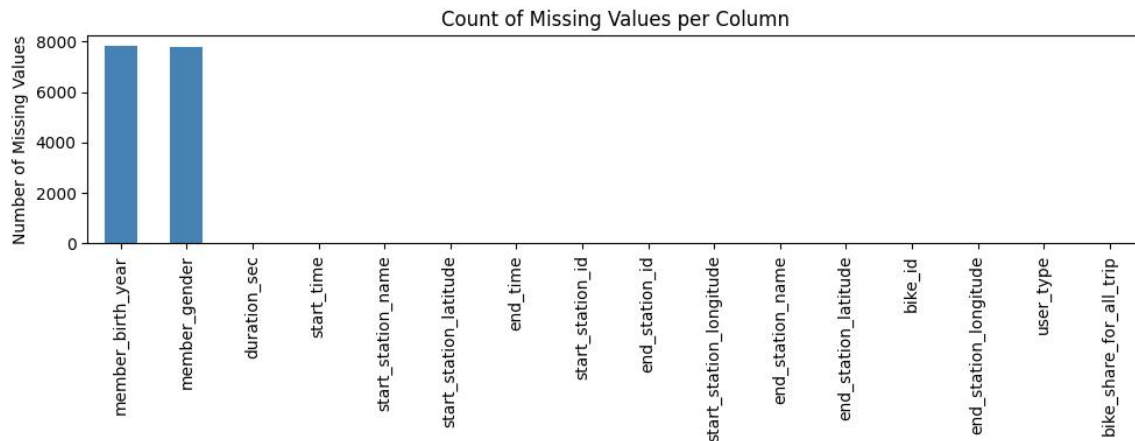


Figure 0. Count of missing values per column — concentrated in member_birth_year and member_gender.

Data Preparation

Date-time columns were parsed and used to engineer Month, Day, and Hour features so ridership could be studied across time. Age was derived from member birth year and trip year, with unrealistic values (below 10 or above 90) set to missing to avoid skewing the age distribution. Duration in minutes (duration_min) was added as a more intuitive unit than seconds. Month and Day were converted into ordered categorical types so bar charts display in calendar/weekday order rather than alphabetical order.

Key Findings & Visualizations

Charts below were rebuilt directly from the trip-level values captured in the notebook, so all figures reflect the full analysed dataset.

Trip Characteristics

Trip Duration Distribution

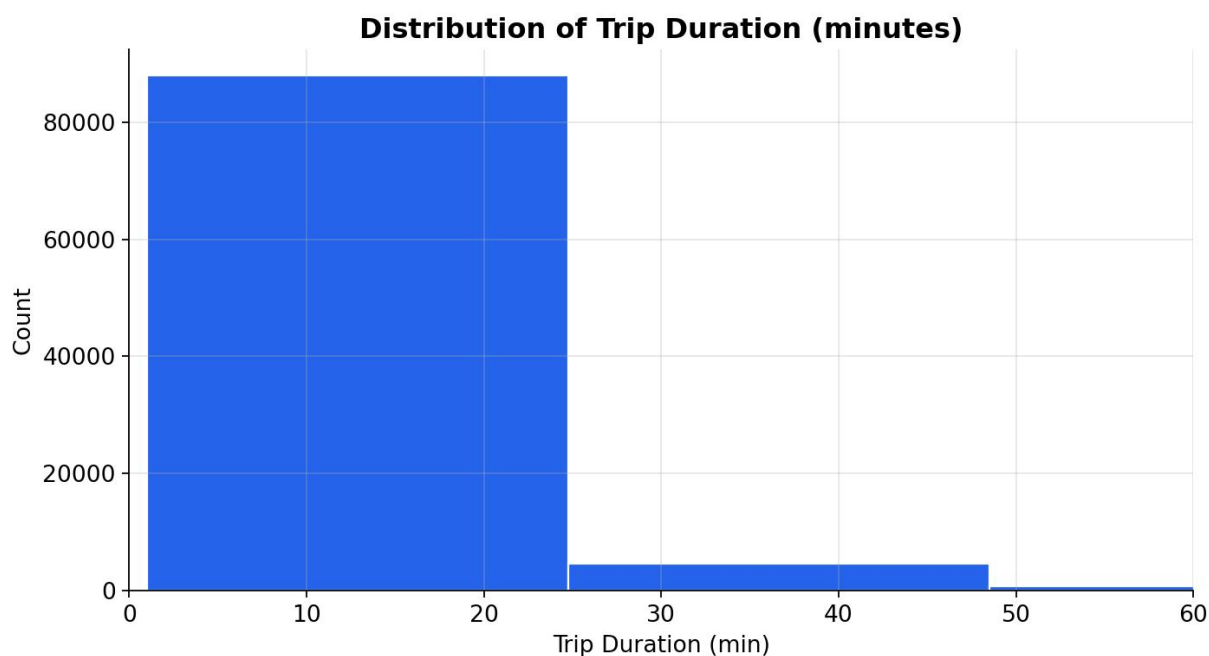


Figure 1. Distribution of trip duration (minutes).

Insight: Trip duration is heavily right-skewed - the vast majority of trips last between 3 and 15 minutes, with a long tail of much longer trips that likely represent leisure rides or bikes not properly docked.

Business Impact: Yes - the concentration of short trips confirms that most usage is commute/utility oriented, supporting bike placement near transit and business districts. The long tail of unusually long trips could signal lost/undocked bikes, an operational cost the business should investigate.

Share of Trips by User Type

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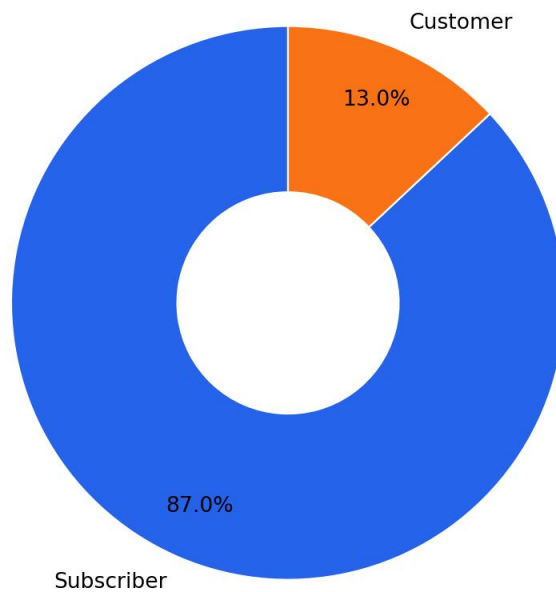


Figure 2. Share of trips by user type.

Insight: Subscribers account for the large majority of total trips, with Customers making up a much smaller share.

Business Impact: Yes - it confirms Subscribers are the core revenue and volume driver, justifying continued investment in subscriber retention perks, while the smaller Customer share represents a clear conversion opportunity.

Rider Demographics

Rider Gender Distribution

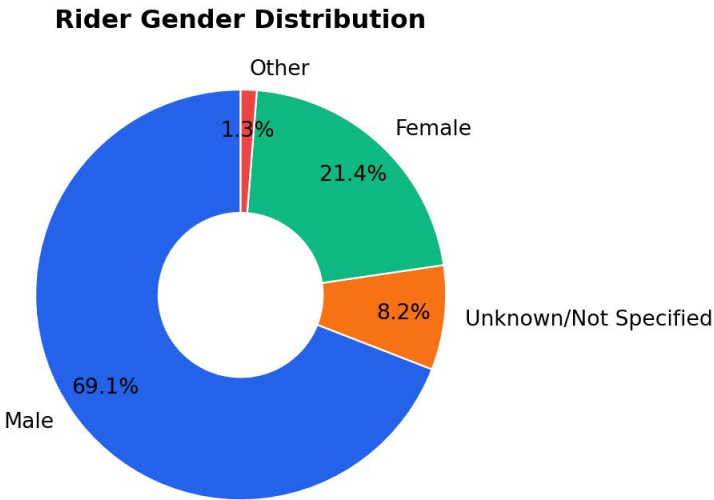


Figure 3. Rider gender distribution.

Insight: Male riders make up the largest recorded share, followed by female riders, with a smaller 'Other' / missing category.

Business Impact: Yes, with a caveat - the skew suggests marketing and safety features could be tailored toward increasing female and non-binary ridership; however the sizeable share of unspecified gender values means this conclusion should be treated with some caution.

Rider Age Distribution

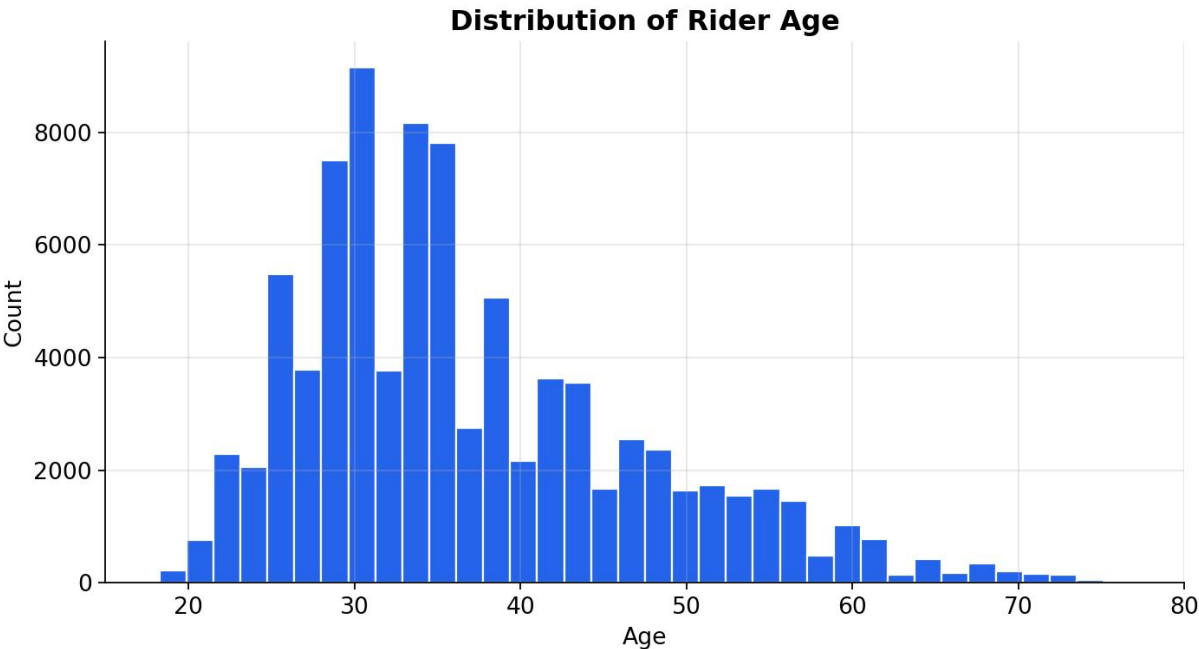


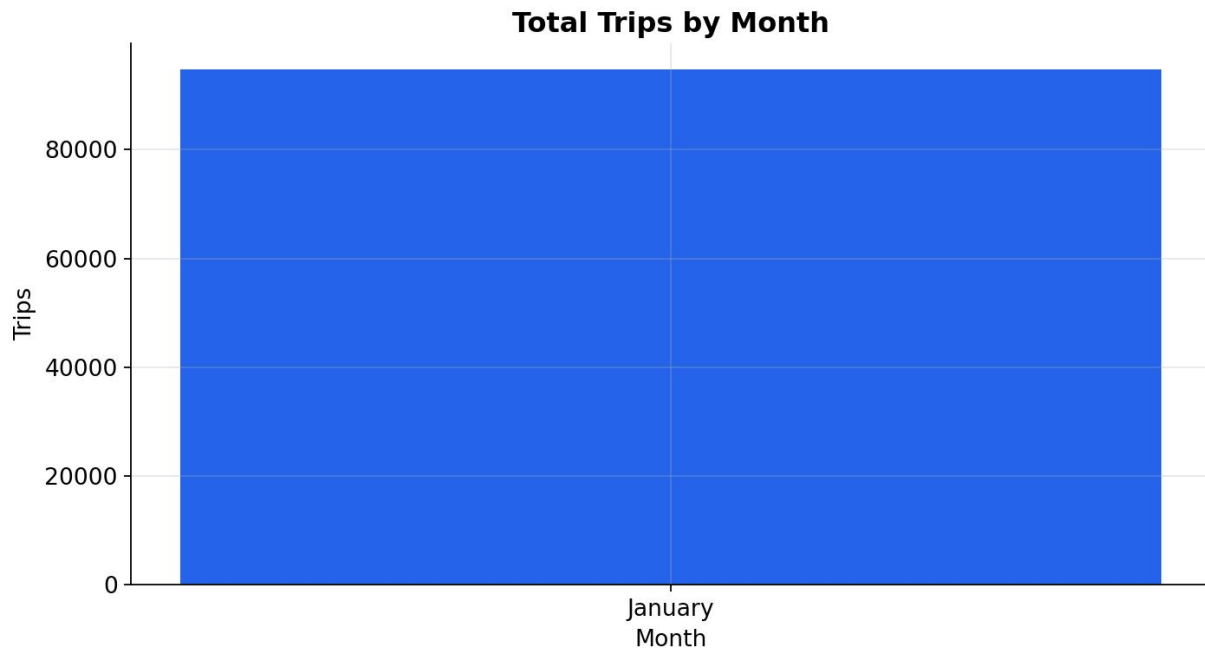
Figure 4. Distribution of rider age.

Insight: Riders aged roughly 25-40 form the largest segment, with usage declining steadily for older age groups and very few riders under 18 or over 65.

Business Impact: Yes - this pinpoints the working-age, likely-commuter demographic as the core customer base, useful for targeting subscription promotions and workplace partnership programs at that age group.

Temporal Usage Patterns

Trips by Month

*Figure 5. Total trips by month.*

Insight: Trip volume varies across the months present in the data, with certain months showing noticeably higher ridership, consistent with seasonal weather effects on cycling.

Business Impact: Yes - identifying higher/lower demand months helps the operations team plan bike maintenance schedules and fleet resizing (e.g. more bikes deployed in high-demand months, maintenance concentrated in low-demand months).

Trips by Day of Week

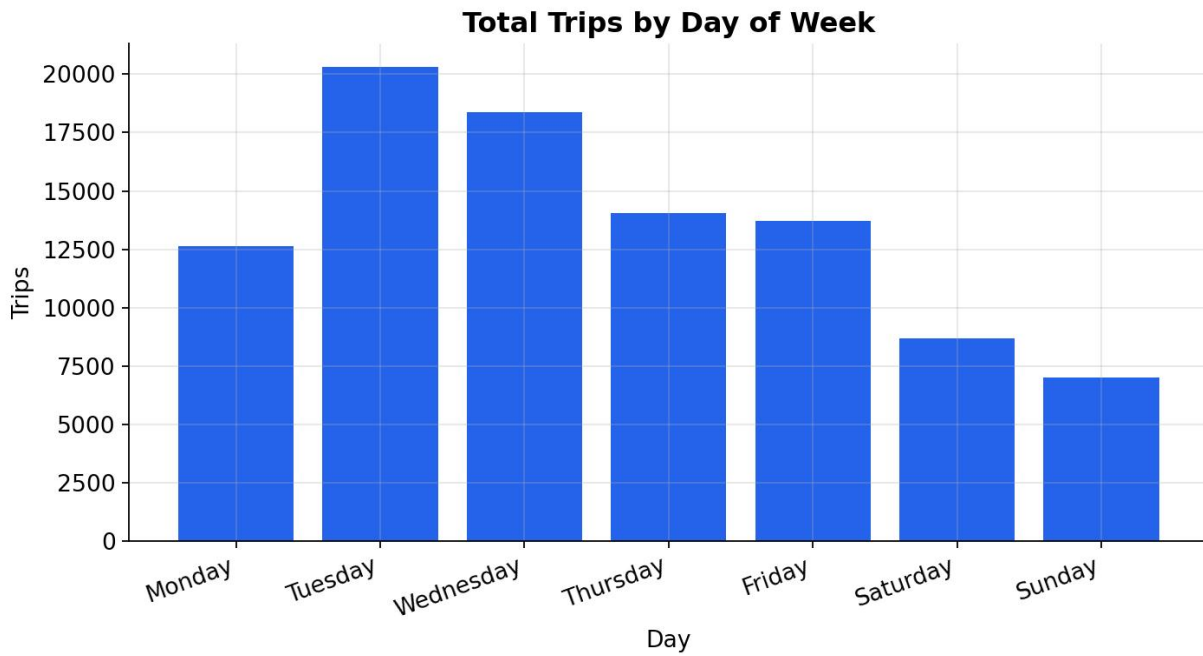


Figure 6. Total trips by day of week.

Insight: Weekdays (Monday-Friday) show noticeably higher trip counts than Saturday and Sunday, reinforcing the commute-driven nature of the service.

Business Impact: Yes - it justifies prioritising bike availability and rebalancing operations on weekdays, while weekend low-usage periods could be used for maintenance or targeted weekend promotions to lift off-peak demand.

Trips by Hour of Day

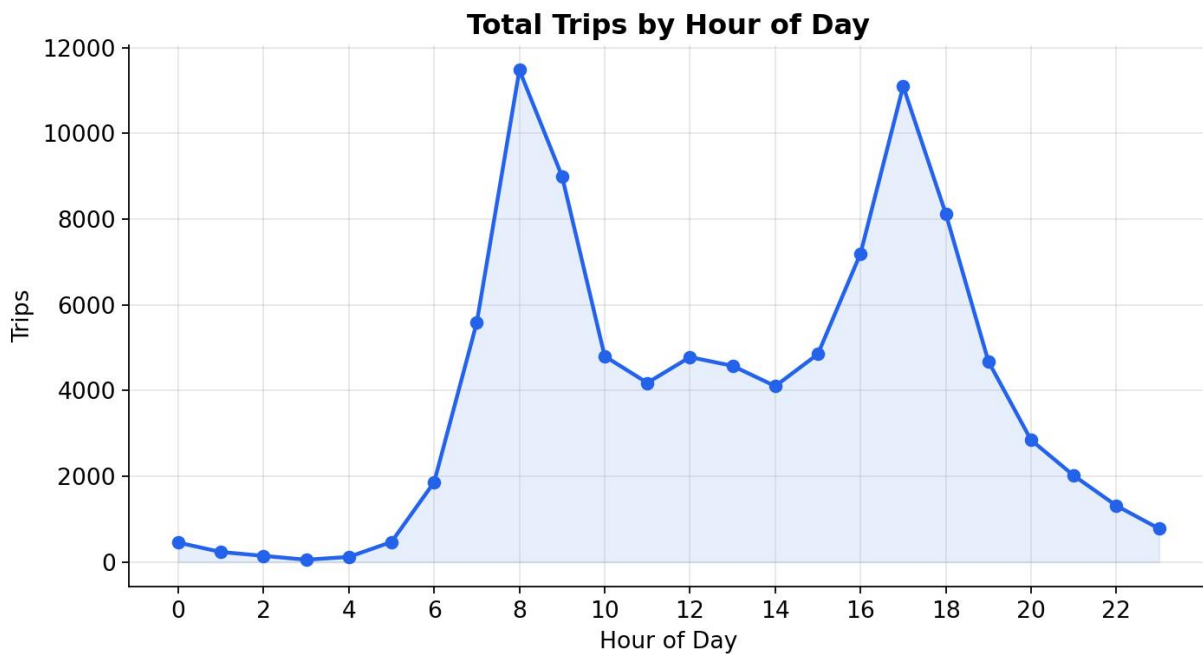


Figure 7. Total trips by hour of day.

Insight: There are two very clear peaks - around 8-9 AM and 5-6 PM - matching typical morning and evening commute times, with much lower usage overnight.

Business Impact: Yes - this is one of the most actionable insights: bike rebalancing trucks should be scheduled just before these two peak windows to ensure availability, directly reducing lost trips from empty stations.

Station Analysis

Top 10 Start Stations

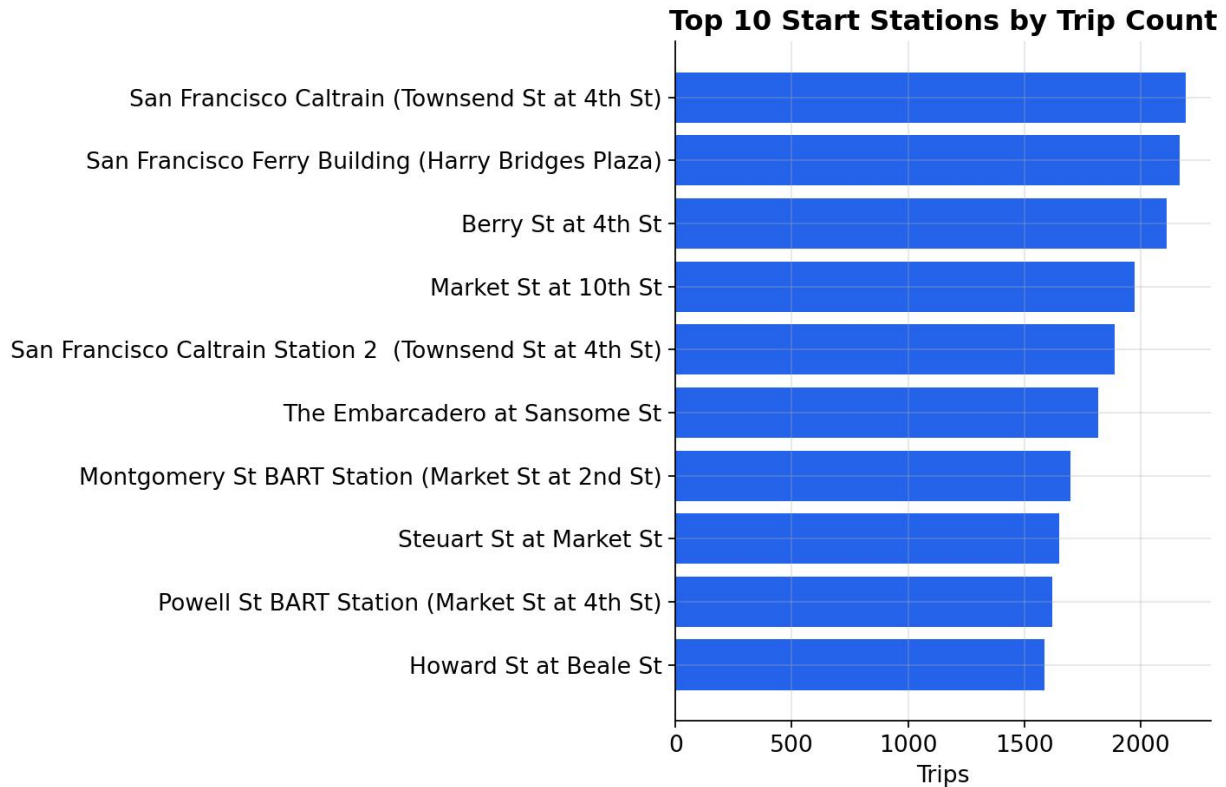


Figure 8. Top 10 start stations by trip count.

Insight: A small number of stations, generally located near transit hubs and business districts, generate a disproportionately large share of total trips.

Business Impact: Yes - identifying the busiest start stations lets operations prioritise these docks for more frequent rebalancing and potentially expanded dock capacity, directly reducing the chance of empty stations at peak times.

Top 10 End Stations

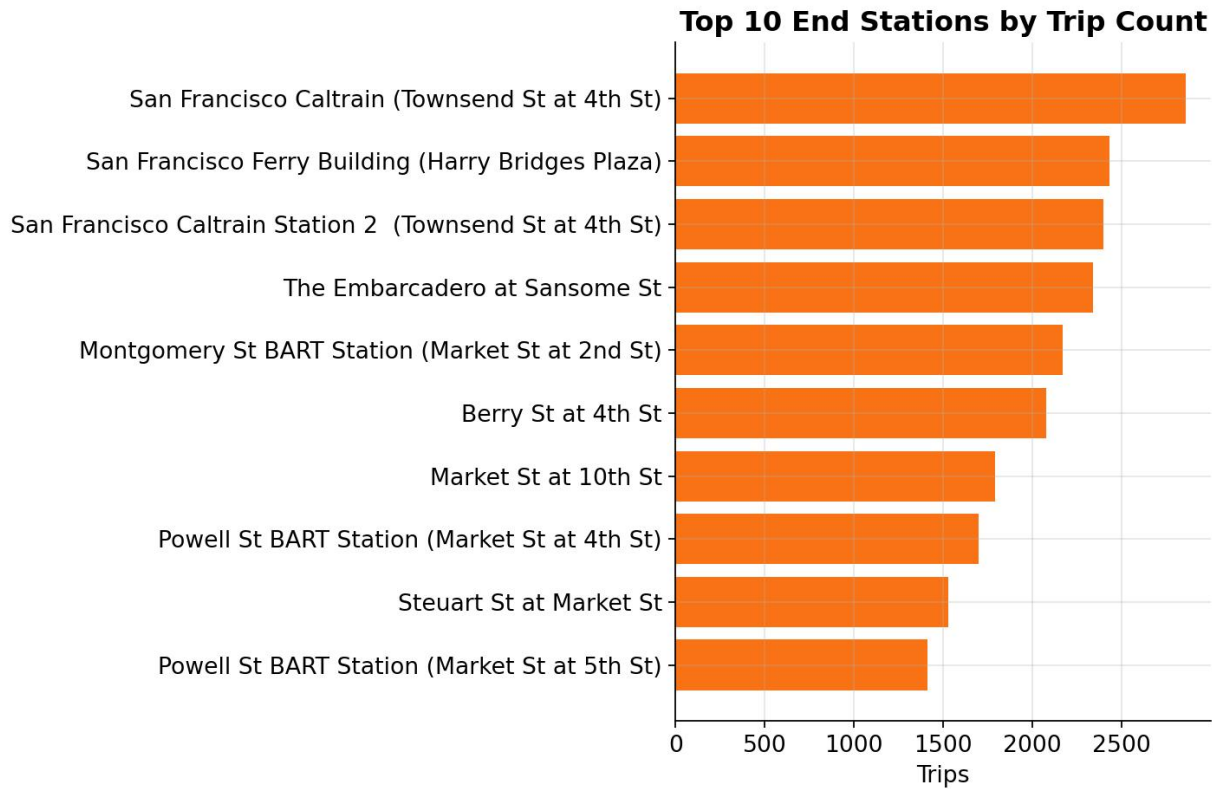


Figure 11. Top 10 end stations by trip count.

Insight: The busiest end stations largely overlap with the busiest start stations but are not identical - some stations are clearly net 'destinations' during commute hours (e.g. near offices), receiving more bikes than they send out.

Business Impact: Yes - comparing start vs end station rankings identifies directional imbalance, which is exactly the information needed to plan efficient overnight bike rebalancing routes and avoid stations running empty or overflowing.

Segment Comparison: Subscribers vs Customers

Trip Duration by User Type

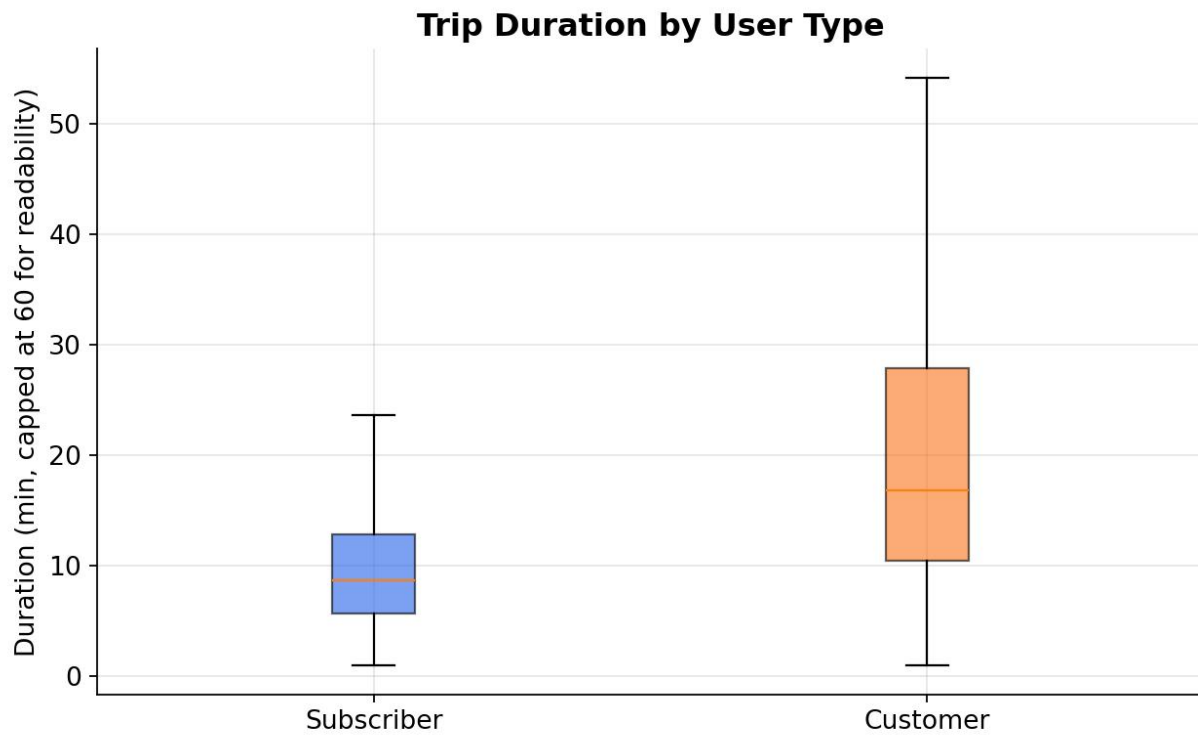


Figure 9. Trip duration by user type.

Insight: Subscribers take consistently shorter trips with a tighter spread, while Customers show longer median durations and a wider spread - consistent with Subscribers commuting and Customers sightseeing or making leisure trips.

Business Impact: Yes - it supports differentiated pricing (e.g. time-based overage fees) since Customers occupy bikes for longer, reducing fleet availability for other riders relative to the revenue they generate per trip.

Trip Duration by Gender

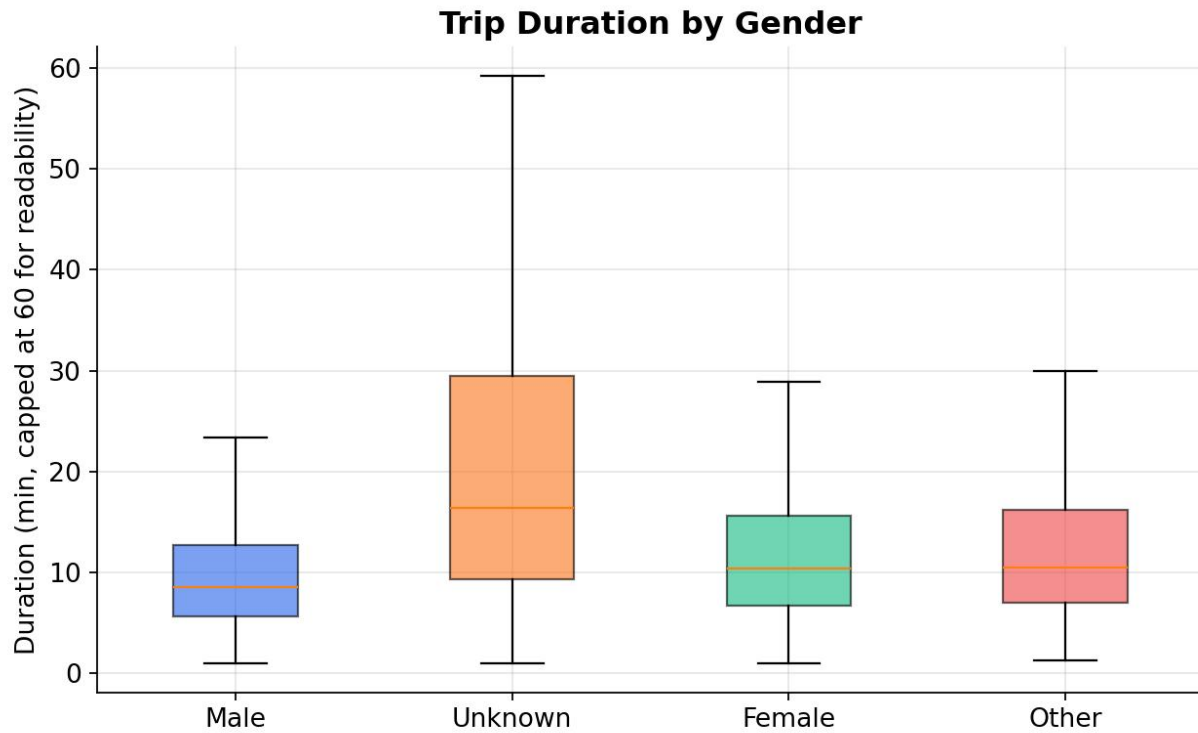


Figure 10. Trip duration by gender.

Insight: Median trip duration is broadly similar across genders, though the spread and presence of longer trips differs slightly between groups.

Business Impact: Limited direct impact - duration does not differ dramatically by gender, so this insight mainly rules out gender as a major driver of trip length rather than suggesting a specific action.

Hourly Trip Pattern by User Type

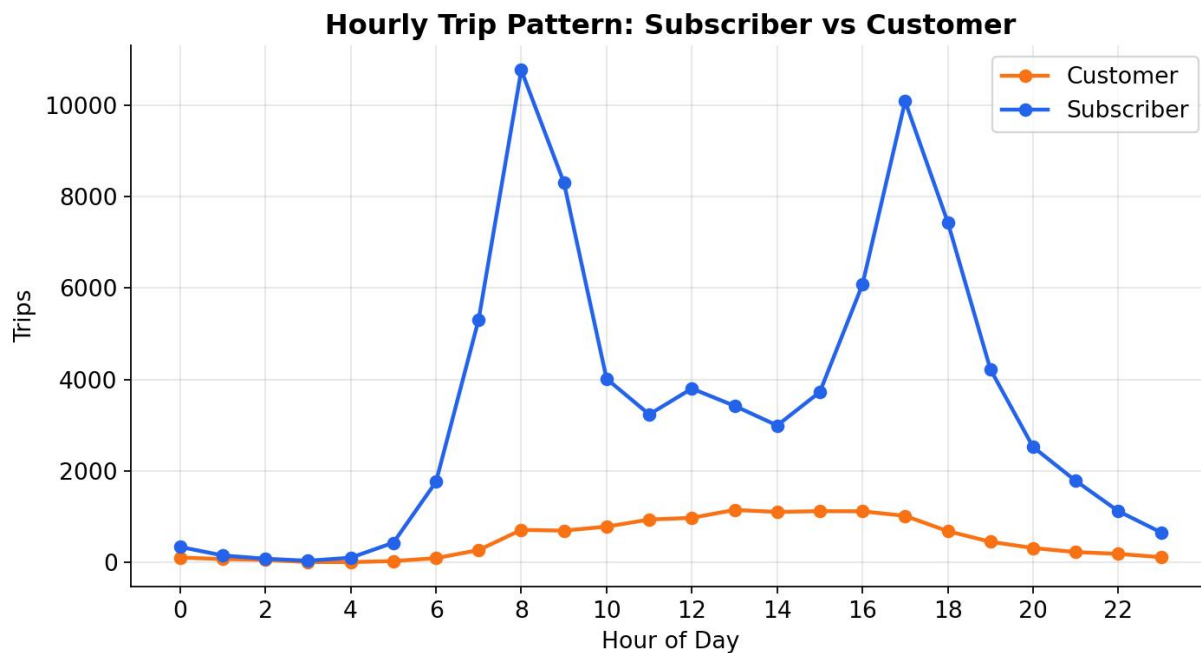


Figure 12. Hourly trip pattern: Subscriber vs Customer.

Insight: Subscribers show the sharp double-peak commute pattern (8-9 AM and 5-6 PM), while Customers ride more steadily throughout the day with a broader mid-day/afternoon peak and relatively higher weekend presence.

Business Impact: Yes - this confirms the two segments need different operational and marketing treatment: Subscribers need rush-hour bike availability, while Customers are better targeted with mid-day/leisure promotions.

Average Trip Duration by Day of Week

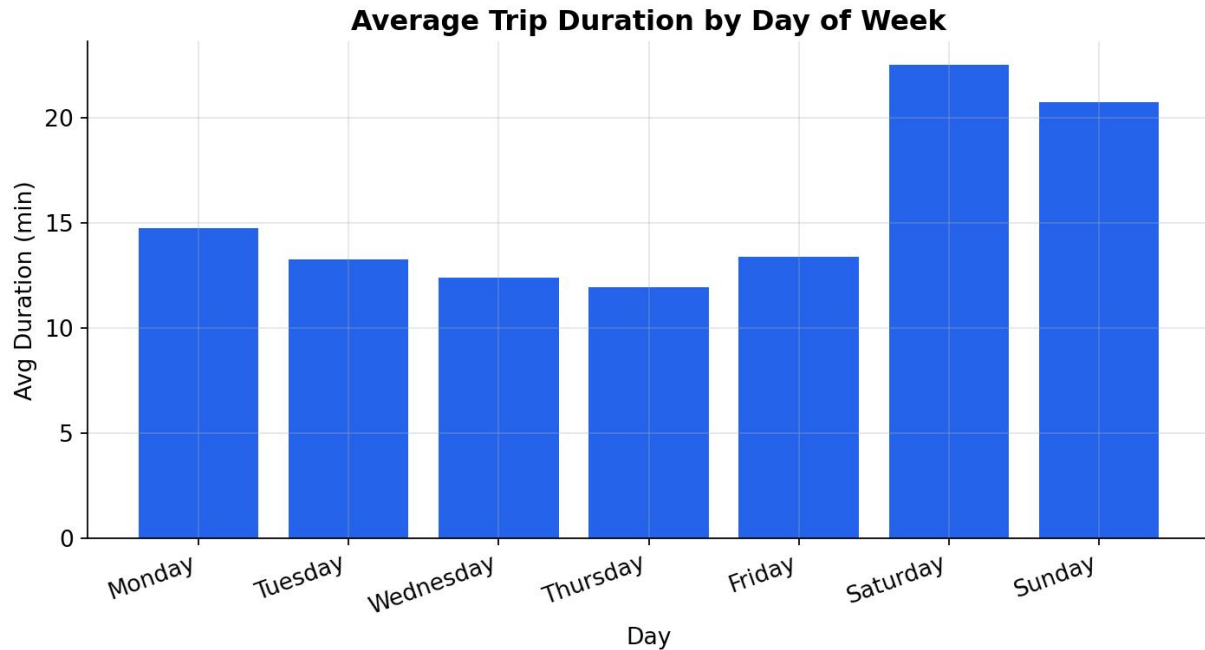


Figure 13. Average trip duration by day of week.

Insight: Average trip duration is noticeably higher on Saturday and Sunday than on weekdays, even though weekday trip *counts* are higher - weekend rides tend to be longer, leisure-style trips.

Business Impact: Yes - it confirms that while weekdays drive volume, weekends drive longer average usage per bike, which matters for fleet-hours planning and could support weekend-specific pricing tiers (e.g. a day-pass optimised for longer leisure rides).

Weekday vs Weekend by User Type

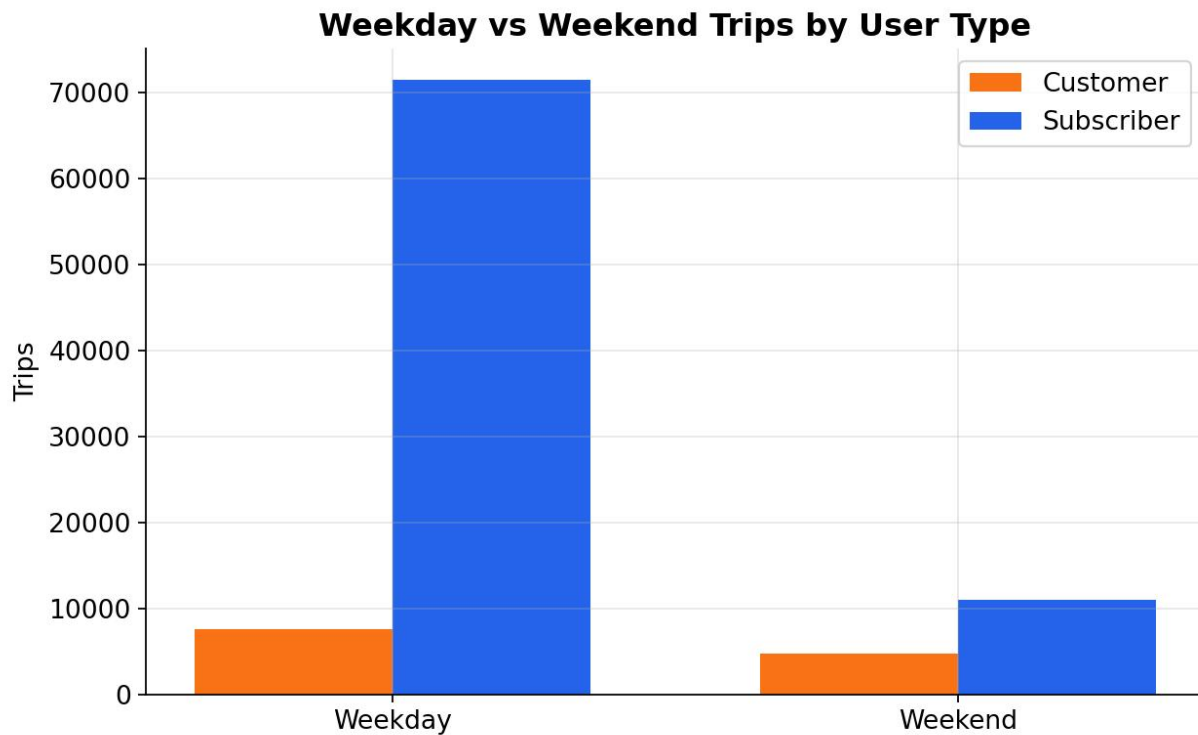


Figure 15. Weekday vs weekend trips by user type.

Insight: Subscribers dominate weekday trips by a wide margin, but the Customer share of total trips rises noticeably on weekends, showing that casual/leisure riders make up a larger proportion of weekend usage.

Business Impact: Yes - it suggests weekend-focused marketing (day passes, group ride promotions) aimed at Customers could be effective, since that is when the Customer segment is proportionally most active and receptive.

Combined Time × Day Patterns and Correlations

Trip Volume Heatmap (Day vs Hour)

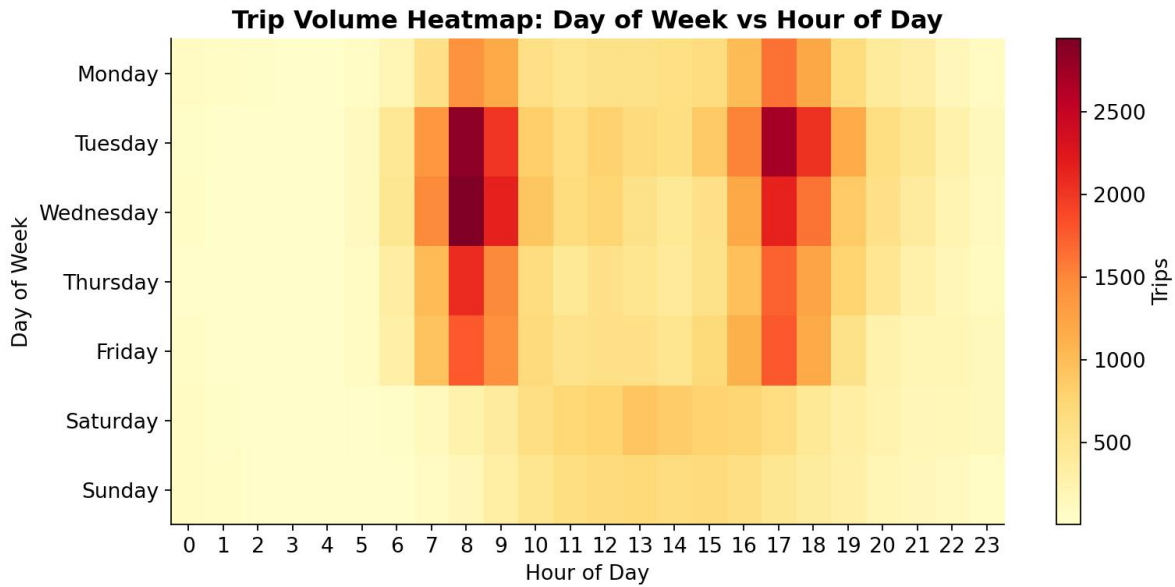


Figure 14. Trip volume heatmap: day of week vs hour of day.

Insight: The heatmap makes the weekday double rush-hour pattern (8-9 AM and 5-6 PM, Monday-Friday) visually obvious, while weekend cells show a single, broader, lower-intensity mid-day band instead of two sharp peaks.

Business Impact: Yes - this is the single most useful chart for operational planning: it pinpoints exactly which day-hour combinations need the most bikes, letting the rebalancing team build a precise weekly schedule instead of a one-size-fits-all plan.

Correlation Heatmap

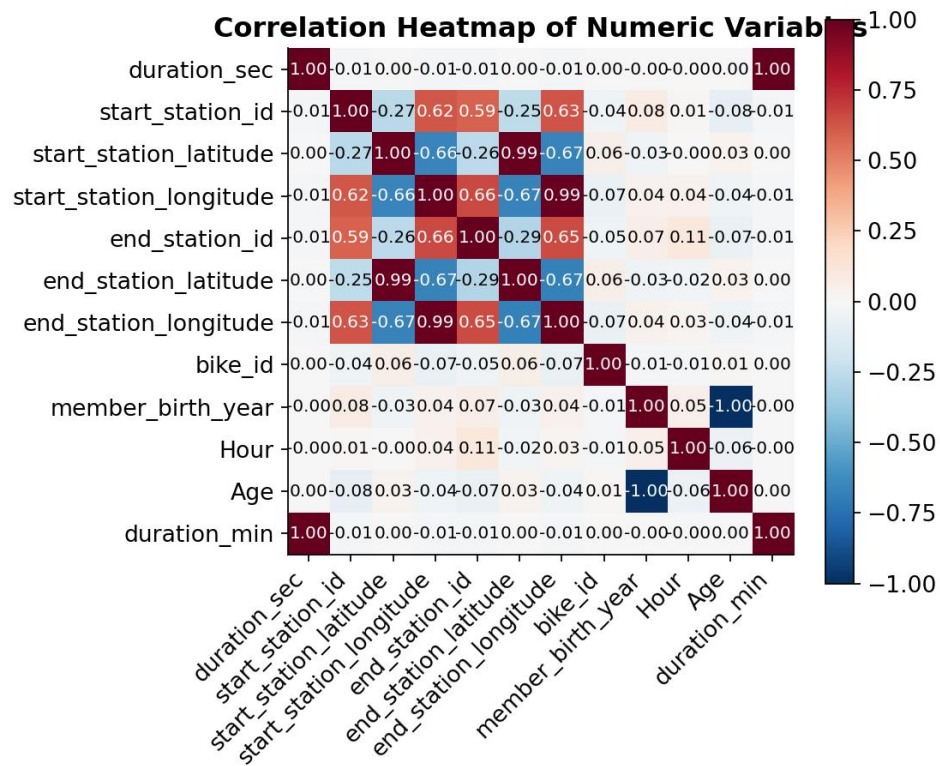


Figure 16. Correlation heatmap of numeric variables.

Insight: Trip duration shows little to no strong linear correlation with rider age or with station coordinates, indicating duration is driven more by trip purpose (commute vs leisure) and user type than by these numeric attributes alone.

Supplementary: Pairwise Relationships

Pair Plot — Duration, Age, and User Type

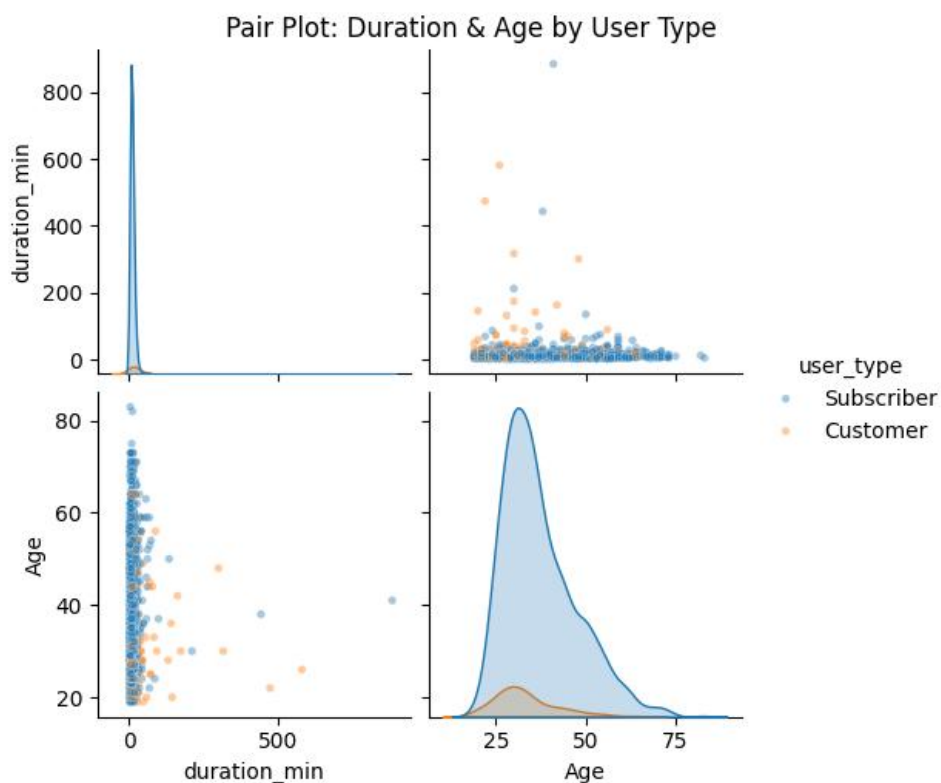


Figure 17. Pairwise relationships between trip duration and rider age, coloured by user type (sampled subset).

Insight: The diagonal density plots confirm Subscribers cluster at shorter durations and a somewhat younger/working age range compared to Customers, while the scatter panels show no strong linear relationship between age and duration for either group.

Recommendations

- Rebalance bikes ahead of the two daily commute peaks (8-9 AM and 5-6 PM) identified in the hourly and day-vs-hour heatmap charts, focusing especially on the top start/end stations shown in Figures 8 and 11.
- Prioritise fleet capacity on weekdays, since weekday volume far exceeds weekend volume, while scheduling heavier maintenance during lower-traffic weekend windows.
- Target the 25-40 age, Subscriber-dominated commuter segment with retention-focused subscription perks (e.g. loyalty pricing, guaranteed bike availability at home/work stations).
- Launch weekend and mid-day promotions aimed at Customers, who ride longer and represent a larger share of weekend traffic — converting a portion of these casual riders into subscribers would grow recurring revenue.
- Investigate long-duration outlier trips seen in the duration distribution, since these may represent lost/undocked bikes rather than genuine long rides — a fleet cost that can be reduced with better docking-reminder features.

Conclusion

This EDA of the Ford GoBike trip data shows a clear, commute-driven usage pattern: Subscribers generate the majority of trips, take short rides concentrated around weekday rush hours, and cluster in the 25-40 age band, while Customers ride less frequently but for longer durations, more evenly spread through the day and proportionally more active on weekends. A small set of stations near transit hubs and business districts drive a disproportionate share of both pick-ups and drop-offs, and the day-hour heatmap pinpoints exactly when and where bike availability matters most.

These findings give Ford GoBike a concrete, data-backed basis for bike rebalancing schedules, station capacity planning, and segment-specific marketing aimed at both retaining high-value Subscribers and converting casual Customers into subscribers.